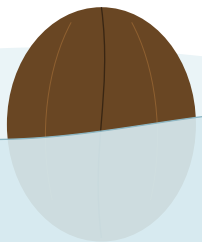
THE MISSING TEXTBOOKS

Dissolving Problems

A Casebook of Theory-Building

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Part I

Building the Right Object

The first family of moves changes what *we talk about: we complete a space so that limits stop escaping, or we axiomatize a notion so that its hidden parameters must be declared.*

1 The Needle and the Noodle

Buffon's problem dissolved by generalization

THE MOVE — GENERALIZE TO SIMPLIFY

IN THIS CHAPTER. Buffon asked for the probability that a needle dropped on ruled paper crosses a line. The classical answer is an integral. Barbier's answer is a sentence: bend the needle into a noodle, and the question becomes so general that it can only have one answer.

1.1 Before: an integral over angles

Rule the plane with parallel lines at distance d and drop a needle of length $\ell \le d$ at random.[1] The classical solution integrates over the angle θ between the needle and the lines:

$$\Pr[\text{cross}] = \frac{2}{\pi}\int_0^{\pi/2} \frac{\ell \sin\theta}{d}\,\mathrm{d}\theta = \frac{2\ell}{\pi d}.$$

The computation is short, but it explains nothing. Why π? Why is the answer linear in ℓ? Change the needle to a bent wire and the integral becomes a different integral.

[1] "At random" means that the distance from the centre of the needle to the nearest line is uniform on $[0, d/2]$, the angle is uniform on $[0, \pi/2]$, and the two are independent.

1.2 The move: let the needle bend

Definition 1.1 (Noodle). A *noodle* is a rigid rectifiable curve of finite length L, dropped with a uniformly random position and orientation.

Let $N(C)$ count the crossings of a noodle C with the lines. The question "what is $\mathbb{E}N(C)$?" sounds harder than Buffon's. It is easier, because it has more structure.

1.3 After: one line of reasoning

Theorem 1.2 (Barbier (1860)). *Every noodle of length L satisfies $\mathbb{E}N(C) = 2L/(\pi d)$.*

Proof. The plan is to show that $\mathbb{E}N$ is proportional to length and then to calibrate the constant with a circle. Cut the noodle into pieces; crossings add, so linearity of expectation makes $\mathbb{E}N$ additive over pieces, even though the pieces move together. An additive, monotone function of length is linear: $\mathbb{E}N(C) = cL$. A circle of diameter d meets the lines exactly twice with probability one, and its length is πd, so $c \cdot \pi d = 2$. ∎

Corollary 1.3. *A needle of length $\ell \le d$ crosses a line with probability $2\ell/(\pi d)$.*

PROOF. A needle with $\ell \le d$ crosses at most one line, so $\Pr[\text{cross}] = \mathbb{E}N$, and Theorem 1.2 applies. ∎

1.4 Costs and limits

WHERE DID THE DIFFICULTY GO? Into linearity of expectation, which holds for dependent pieces, and into the existence of one calibrating shape whose crossing number is constant.

1.5 The move, distilled

THE MOVE · GENERALIZE TO SIMPLIFY

IN ONE SENTENCE. Replace a question about one object by the same question about a family that has more structure; the structure may force the answer.

TRIGGERS. A computation that gives the right number but no reason; an answer that is suspiciously linear.

ANTI-PATTERN. A generalization whose proof is longer than the special case.

DISSOLUTION TEST

SHORTER PROOFS	A trigonometric integral becomes a two-line argument.
FEWER ASSUMPTIONS	Straightness of the needle is no longer used.
UNIFICATION	Needles, polygons and curves obey one formula.
NEW REACH	Expected crossings of every rectifiable curve.

Exercises

1.1 CONSOLIDATION. Prove that an additive monotone function on $[0, \infty)$ is linear.

1.2 TRANSFER. ⋆ Use the same move to compute the expected number of crossings of a random line with a convex curve. *(Hint: Calibrate with a circle again.)*

1.3 RESEARCH. ⋆⋆ Which other classical integrals in geometric probability dissolve in the same way?

Notes and References

The noodle argument is due to Barbier (1860); a modern account is in Aigner and Ziegler (2018). Pólya's "inventor's paradox" (Pólya 1945) is the general principle.

References

Aigner, M. and G. M. Ziegler (2018). *Proofs from THE BOOK*. 6th ed. Berlin: Springer.

Barbier, J.-É. (1860). "Note sur le problème de l'aiguille et le jeu du joint couvert". In: *Journal de Mathématiques Pures et Appliquées*. 2nd ser. 5, pp. 273–286.

Pólya, G. (1945). *How to Solve It*. Princeton, NJ: Princeton University Press.

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